Software Requirements Specification

for

Group Based Movie Recommendation System

Version 1.0 Draft

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Revision History

Name	Date	Reason For Changes	Version
Draft 1	11th October 2026	Initial Draft	1.0

1. Introduction

The Group-Based Movie Recommendation System is designed to simplify the process of choosing a movie that suits the preferences of a group of people. The system allows users to create or join groups, indicate their interest in a selection of movies based on their current mood, and receive a ranked list of recommendations tailored to the group. The recommendation process considers individual preferences, publicly available movie ratings, and feedback from previous sessions of the same group to improve future suggestions. By combining these factors, the system aims to provide relevant recommendations that balance the interests of group members while making shared movie selection more convenient and personalized.

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for the Group-Based Movie Recommendation System Version 1.0. It describes the intended behavior, interfaces, constraints, security objectives, and quality requirements, and provides a reference for design, implementation, testing, and validation.

1.2 Document Conventions

Requirements use unique identifiers: FR for functional requirements, NFR for nonfunctional requirements, SO for security objectives, and SR for security requirements. The term "shall" denotes a mandatory requirement. Each requirement is written to be concise, unambiguous, necessary, and verifiable where practicable. Priority labels are High, Medium, or Low. Numerical targets are proposed acceptance criteria and must be confirmed during project planning.

1.3 Intended Audience and Reading Suggestions

This document will be provided to the following teams:

- *Test Team: will review if requirements are written correctly or not*
- *Customer: to see if customer requirements are captured correctly or not*
- *Project Manager: to prepare project plan and schedule*
- *DepOps Engineers: to provide DevOps tools*

1.4 Product Scope

In this project, the scope involves:

- *Developing a web-based user interface using HTML, CSS, and JavaScript for user registration, login, group creation, movie preference submission, and viewing recommendations.*
- *Developing a group-based movie recommendation system that considers the current preferences of all group members, publicly available movie ratings, and historical feedback from previous sessions of the same group.*
- *Implementing a content-based recommendation approach to estimate group compatibility for movies across the local catalogue, including movies not presented during the initial preference-selection stage.*
- *Integrating a movie dataset containing titles, genres, descriptions, and ratings to support recommendation generation and ranking.*
- *Implementing a feedback mechanism that allows members to rate recommended movies on a scale of 1 to 5 and uses the aggregated feedback to improve recommendations in subsequent sessions.*
- *Evaluating the proposed hybrid recommendation approach against baseline methods using appropriate ranking metrics and group compatibility measures.*

1.5 References

- *Movie datasets containing movie titles, genres, descriptions, and ratings, to be selected from sources such as MovieLens or Kaggle.*
- *Statement of Work from Customer*

2. Overall Description

2.1 Product Perspective

The application is a standalone web system with a presentation layer, application/API layer, recommendation module, and persistent data store. The application retrieves movie metadata and public ratings from a selected local dataset or approved data source. Group preference and feedback records are stored separately from general movie metadata so that a group's historical feedback can be applied only to that group's later sessions.

2.2 Product Functions

- *Register users and authenticate returning users.*
- *Create groups and join groups through a unique code or link.*
- *Enforce unique display names within each group.*
- *Present a configurable, genre-diverse set of movies to each member; the initial target is 10 movies per member.*
- *Collect and lock each member's Interested / Not Interested selections for a recommendation session.*
- *Wait until all participating members submit before generating the session's recommendations.*
- *Estimate compatibility for catalogue movies that were not in a member's initial selection using available movie metadata and submitted preferences.*
- *Rank the catalogue using current group compatibility, public movie ratings, and group-specific historical feedback.*
- *Display ranked recommendations with title, description, and available rating.*
- *Collect 1–5 member ratings, calculate the arithmetic mean, and retain the group's history for later sessions.*
- *Validate inputs, enforce authorization, and report failures without falsely confirming successful operations.*

2.3 User Classes and Characteristics

User Class	Characteristics
Group member	*Primary user; can register/log in, create or join a group, submit session preferences, view recommendations, and rate movies. Expected to have basic web-browsing skills.*
Group creator	*A group member who initiates a group and shares its joining code/link. No separate administrative privileges are assumed unless explicitly implemented.*
Administrator	*Maintains the movie catalogue or application configuration if such capabilities are included. Administrative access must be restricted*

2.4 Operating Environment

- *Client: modern desktop or mobile web browser with JavaScript enabled.*
- *Server: application runtime capable of hosting the web/API layer and recommendation logic*
- *Database: persistent relational or equivalent database capable of storing accounts, groups, sessions, preferences, movie metadata, and ratings*
- *Network: internet or local network connection between browser and application server.*

2.5 Design and Implementation Constraints

- *The initial implementation shall use a local movie catalogue available to the application; a final dataset and licence must be selected before release.*
- *The recommendation formula shall support configurable weights for current compatibility, public rating, and historical group feedback. Weights shall be selected through validation experiments rather than treated as universally optimal constants.*
- *Member-level influence in current group preference aggregation shall be equal.*
- *Public rating values may originate from different scales. The implementation must retain the source and scale and document the normalization method before combining ratings.*
- *A completed session's submitted preferences and generated result shall not be silently changed by feedback collected later.*

2.6 User Documentation

The initial release shall include a concise user guide or in-app help explaining account access, creating/joining a group, submitting preferences, understanding recommendations, and rating movies. Developer documentation shall describe the data schema, recommendation score, configuration values, and evaluation procedure.

2.7 Assumptions and Dependencies

- *Users have access to a supported browser and network connection.*
- *The selected movie dataset contains enough valid titles and metadata to produce genre-diverse selections and catalogue-wide candidates.*
- *Public ratings and descriptions are available for at least a useful subset of movies, and their use complies with the dataset/provider terms.*
- *Historical personalization is group-specific; it does not assume that a group's past ratings represent an individual member's general taste.*
- *A session has a defined set of participating members before preference submission begins.*

3. External Interface Requirements

3.1 User Interfaces

The system shall provide a web-based interface through which users can access account, group, preference, recommendation, and feedback functions. The interface shall present clear navigation and indicate the current session status.

- *The registration and login screens shall provide fields for required credentials and display a clear message when validation or authentication fails.*
- *The group screen shall allow an authenticated user to create a group or enter a joining code/link to join an existing group.*
- *The preference screen shall display the selected movies with available titles, genres, or descriptions and provide explicit "Interested" and "Not Interested" controls.*
- *The submission screen shall indicate whether preferences have been saved and whether other participating members are still required to submit.*
- *The recommendation screen shall display movies in ranked order with each movie's title, description, and available public rating.*
- *The feedback interface shall accept a rating from 1 to 5 and confirm whether the rating was saved.*
- *Error messages shall explain the issue in user-understandable language and shall not expose technical secrets.*

3.2 Hardware Interfaces

The system does not require specialized hardware. Users shall access the application through a computer, tablet, or mobile device with a supported web browser, display, input device, and network connectivity. No direct interface with external hardware is required for the initial release.

3.3 Software Interfaces

- *The browser-based client shall communicate with the application server through documented API endpoints.*
- *The application server shall read and write user accounts, group membership, recommendation sessions, movie metadata, submitted preferences, recommendations, and ratings in a persistent database.*
- *The recommendation module shall receive movie metadata, normalized public ratings, current-session preferences, and eligible historical feedback for the same group. It shall return an ordered list of candidate movies.*
- *The movie-data interface shall provide, where available, movie identifiers, titles, genres, descriptions, public rating values, rating sources, and original rating scales.*
- *The final web framework, database engine, dataset, rating provider, and software versions shall be documented once selected.*

3.4 Communications Interfaces

- *The deployed application shall use HTTPS to protect credentials and private group data during transmission.*
- *Client-server requests and responses shall use a documented structured format, such as JSON, with consistent success and error responses.*
- *The system shall not transmit passwords or authentication tokens in URL query parameters.*
- *The application shall handle a failed or interrupted request by displaying an error and shall not indicate that an operation succeeded unless the server confirms success.*

4. System Features

The functional requirements are organized by the main services provided by the system. Each requirement has a unique identifier to support implementation, testing, and traceability.

4.1 User and Group Management

4.1.1 Description and Priority

Allows users to access the application and establish a shared group context. Priority: High.

4.1.2 Stimulus/Response Sequences

- *Stimulus: a visitor submits valid registration details. Response: the system creates an account or displays a validation error.*
- *Stimulus: an authenticated user creates a group or submits a joining code/link. Response: the system creates the group or adds the user if the request is valid.*

4.1.3 Functional Requirements

- *FR-01: The system shall allow a user to register an account and log in using valid credentials. It shall reject invalid credentials without granting access.*
- *FR-02: The system shall allow an authenticated user to create a group or join an existing group using a unique joining code or link. It shall reject invalid or expired codes or links.*
- *FR-03: The system shall enforce unique display names among members of the same group.*

4.2 Preference Collection and Session Management

4.2.1 Description and Priority

Collects each participating member's current-session preferences and ensures that recommendations are generated from completed submissions. Priority: High.

4.2.2 Stimulus/Response Sequences

- *Stimulus: a member opens the preference screen. Response: the system presents a genre-diverse selection of movies and allows a preference choice for each movie.*
- *Stimulus: a member submits preferences. Response: the system validates and saves the submission, locks it for the session, and waits for remaining participating members.*

4.2.3 Functional Requirements

- *FR-04: The system shall present each participating member with a configurable selection of movies spanning multiple genres, using 10 movies per member as the initial target, subject to catalogue availability.*
- *FR-05: The system shall allow each member to mark every presented movie as either "Interested" or "Not Interested."*
- *FR-06: The system shall store each submitted preference with the corresponding user, group, movie, and recommendation-session identifiers.*
- *FR-07: The system shall lock a member's submitted preferences for the current session and shall generate recommendations only after all participating members have submitted their preferences.*

4.3 Recommendation Generation and Ranking

4.3.1 Description and Priority

Ranks candidate movies using current group compatibility, public ratings, and historical feedback belonging to the same group. Priority: High.

4.3.2 Stimulus/Response Sequences

- *Stimulus: all participating members submit preferences. Response: the system evaluates eligible catalogue movies and generates a ranked recommendation list.*
- *Stimulus: a member opens the result screen. Response: the system displays the ranked list and available movie information.*

4.3.3 Functional Requirements

- *FR-08: The system shall aggregate current preferences from all participating members with equal member-level influence when calculating group compatibility.*
- *FR-09: The system shall evaluate candidate movies from the entire local catalogue, including movies not presented during preference collection. For unevaluated candidates, it shall estimate compatibility from available movie metadata and submitted preferences.*
- *FR-10: The system shall rank candidate movies using current group compatibility, public movie ratings, and available historical feedback from the same group. The scoring method shall support configurable weights. If a group has no historical feedback, the system shall rank using current preferences and available public ratings.*
- *FR-11: The system shall display a ranked list of recommended movies containing each movie's title, description, and available public rating.*

4.4 Feedback and Historical Personalization

4.4.1 Description and Priority

Collects individual member ratings and uses group-specific feedback to influence recommendations in subsequent sessions. Priority: High.

4.4.2 Stimulus/Response Sequences

- *Stimulus: a member submits a rating for a recommended movie. Response: the system validates and stores the rating and updates the group's aggregate rating.*
- *Stimulus: a later session requests recommendations. Response: the system retrieves eligible historical feedback belonging to that group.*

4.4.3 Functional Requirements

- *FR-12: The system shall allow each group member to rate a recommended movie using an integer rating from 1 to 5, inclusive.*
- *FR-13: The system shall store each member's rating separately and calculate the arithmetic mean of the available member ratings for the group and movie.*
- *FR-14: The system shall retrieve historical feedback belonging to the same group and use it to influence recommendations in subsequent sessions without modifying preferences or results of a completed session.*

4.5 Data Validation and Error Handling

4.5.1 Description and Priority

Ensures that invalid operations are rejected and that users receive accurate status information. Priority: High.

4.5.2 Stimulus/Response Sequences

- *Stimulus: a user submits missing, malformed, out-of-range, or unauthorized data. Response: the system rejects the operation, displays an understandable error, and preserves previously saved data.*

4.5.3 Functional Requirements

- *FR-15: The system shall validate user inputs and preserve successfully saved data. If an operation fails, it shall display an understandable error message and shall not report the unsuccessful operation as successful.*

5. Other Nonfunctional Requirements

5.1 Performance Requirements

NFR-01: Under a test workload with a catalogue of up to 10,000 movies and groups of up to 10 members, the system shall return a ranked recommendation list within 3 seconds for at least 95% of recommendation requests, excluding external network delays for data sources not controlled by the system.

Verification: the team shall run a documented performance test against a representative deployment and record request latency and failure rate. The target is a proposed acceptance criterion and should be confirmed against the available hardware.

5.2 Safety Requirements

The system does not control physical equipment or provide safety-critical advice. The main foreseeable risks are loss of user data, exposure of private group activity, and misleading presentation of incomplete movie ratings.

- *The system shall identify missing public ratings as unavailable rather than generating or displaying fabricated values.*
- *The system shall not represent a recommendation as guaranteed to satisfy every group member.*
- *The system shall preserve saved preferences and feedback when recommendation generation fails.*

5.3 Security Requirements

5.3.1 Security Objectives

- *SO-01 (Confidentiality): Only authenticated and authorized users shall access private group information, including membership details, submitted preferences, and individual feedback.*
- *SO-02 (Integrity): The system shall prevent unauthorized modification of accounts, group membership, submitted preferences, session results, and ratings.*
- *SO-03 (Secure Access): Authentication and session-management mechanisms shall prevent unauthenticated users from accessing protected application functions.*

5.3.2 Security Requirements

- *SR-01: The system shall require authentication before allowing a user to access protected group, preference, recommendation-history, or feedback data.*
- *SR-02: For every protected group operation, the system shall verify that the authenticated user is a member of the relevant group or has an explicitly authorized administrative role.*
- *SR-03: The system shall store passwords only as salted, one-way password hashes generated by a recognized password-hashing algorithm; plaintext passwords shall not be stored.*
- *SR-04: The system shall validate user-supplied input on the server and use parameterized database queries or an equivalent safe data-access mechanism to reduce injection risks.*
- *SR-05: A deployed system shall transmit credentials and protected user/group data over HTTPS.*
- *SR-06: Error responses shall not disclose passwords, authentication tokens, internal stack traces, database credentials, or other secrets.*

Verification: test unauthenticated access, cross-group access attempts, password storage, invalid inputs, transport configuration, and error responses.

5.4 Software Quality Attributes

- *NFR-02 (Reliability and Data Integrity): The system shall preserve saved preferences, group membership, session results, and ratings across application restarts. Stored records shall reference valid related records.*
- *NFR-03 (Privacy): The system shall expose individual preferences and ratings only to authorized users under the defined group-visibility policy and shall collect only information needed for account, group, and recommendation functions.*
- *NFR-04 (Usability): In a usability test with at least five first-time participants, at least four shall complete the core workflow (create/join a group, submit preferences, view recommendations, and submit a rating) without assistance.*
- *NFR-05 (Maintainability and Testability): The implementation shall separate group management, preference/session management, movie-data access, recommendation scoring, and feedback handling into independently testable modules.*
- *NFR-06 (Recommendation Quality): The team shall evaluate the hybrid recommender against at least a public-rating-only baseline and a current-preference-only baseline using held-out data. The evaluation shall report NDCG@10 or another justified ranking metric and a group-compatibility/satisfaction measure where suitable feedback data is available.*
- *NFR-07 (Robustness): The system shall handle missing movie descriptions or public ratings without terminating recommendation generation; unavailable values shall be marked or omitted rather than fabricated.*

5.5 Business Rules

- *BR-01: Each participating group member shall have equal member-level influence in current-session compatibility aggregation.*
- *BR-02: "Interested" and "Not Interested" shall represent a member's current-session preference and shall not automatically be treated as permanent individual likes or dislikes.*
- *BR-03: Submitted preferences shall be locked for that session. Feedback from later sessions may influence future rankings but shall not rewrite a completed session.*
- *BR-04: Individual movie ratings shall be integers from 1 to 5. The group aggregate shall be the arithmetic mean of available individual ratings for that group and movie.*
- *BR-05: Public ratings from different sources or scales shall not be combined until normalized using a documented method. The original source and scale shall be retained.*
- *BR-06: Recommendation weights shall be configurable and selected through validation experiments; no particular weight combination is assumed to be optimal.*

6. Other Requirements

6.1 Data Requirements

The persistent data model shall represent users, groups, group memberships, recommendation sessions, movie records, session preferences, generated recommendations, and individual ratings. Records shall contain identifiers that associate preferences and feedback with the correct user, group, movie, and session. Data retention and account deletion rules remain to be finalized.

6.2 Recommendation Evaluation

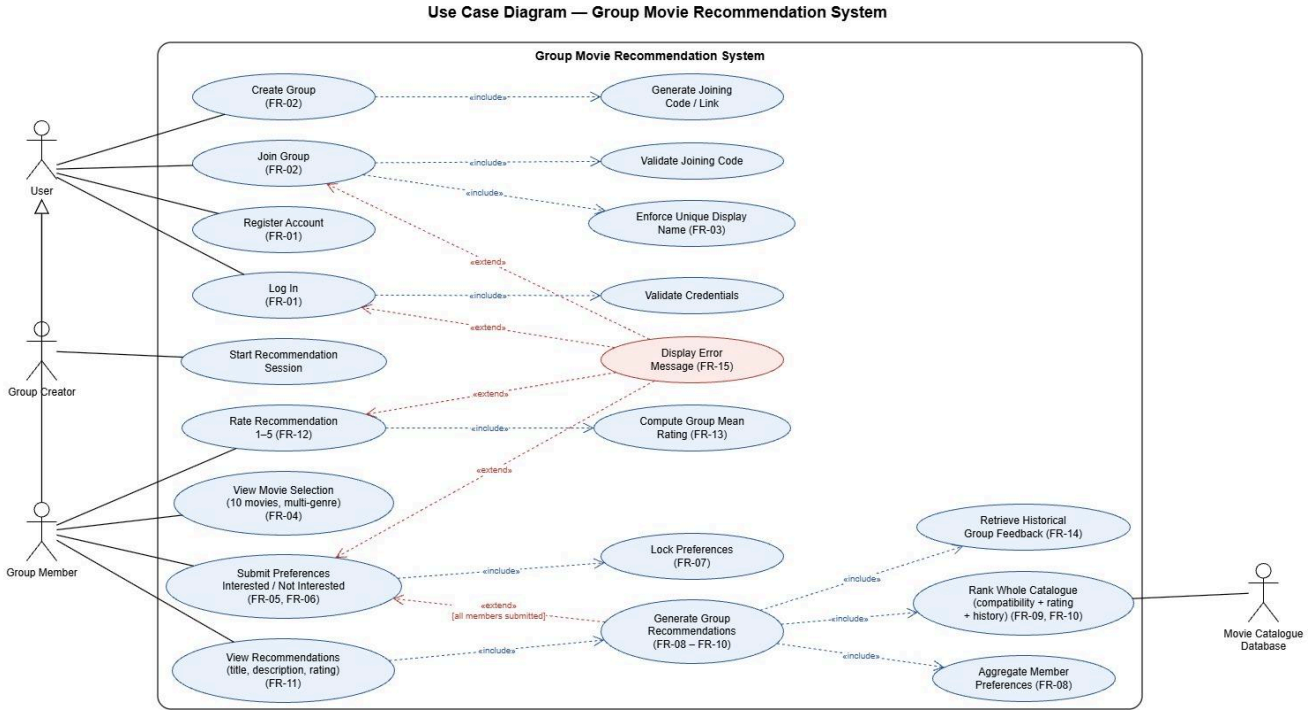
The evaluation shall compare at least three methods: public-rating-only ranking, current-preference-only ranking, and the proposed hybrid ranking. Evaluation shall use held-out data and shall avoid using the same feedback both to generate a recommendation and to treat that recommendation as independently validated ground truth. The selected metrics and data-splitting protocol shall be documented. Collaborative filtering is optional for the initial release.

Appendix A: Glossary

- *Candidate movie: A movie considered for ranking, including movies not shown during preference collection.*
- *Current-session preference: A member's Interested / Not Interested response for the active recommendation session.*
- *Group compatibility: A score estimating how well a candidate movie fits the current preferences of participating group members.*
- *Public rating: A rating supplied by a selected dataset or rating source, with its original scale and source.*
- *Historical group feedback: Ratings or feedback from earlier sessions of the same group.*
- *SRS: Software Requirements Specification.*

Appendix B: Analysis Models

B.1 UML Use Case Diagram



Appendix C: To Be Determined List

- *TBD-01: Select the final movie dataset*
- *TBD-02: Select the public rating source(s) and document normalization for different rating scales.*
- *TBD-03: Confirm the web framework, database engine, deployment environment, and supported browser versions.*
- *TBD-05: Define the procedure for a member who leaves, does not submit preferences, or does not rate recommendations.*
- *TBD-06: Choose the initial preference-selection algorithm and the rule for excluding or penalizing movies marked Not Interested.*
- *TBD-07: Confirm the performance test environment and acceptance target.*
- *TBD-09: Confirm administrator role scope*
- *TBD-10: Define data retention and account deletion policy.*

End of SRS.